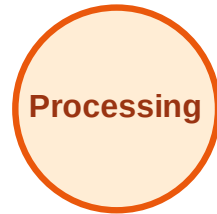


BFS Traversal

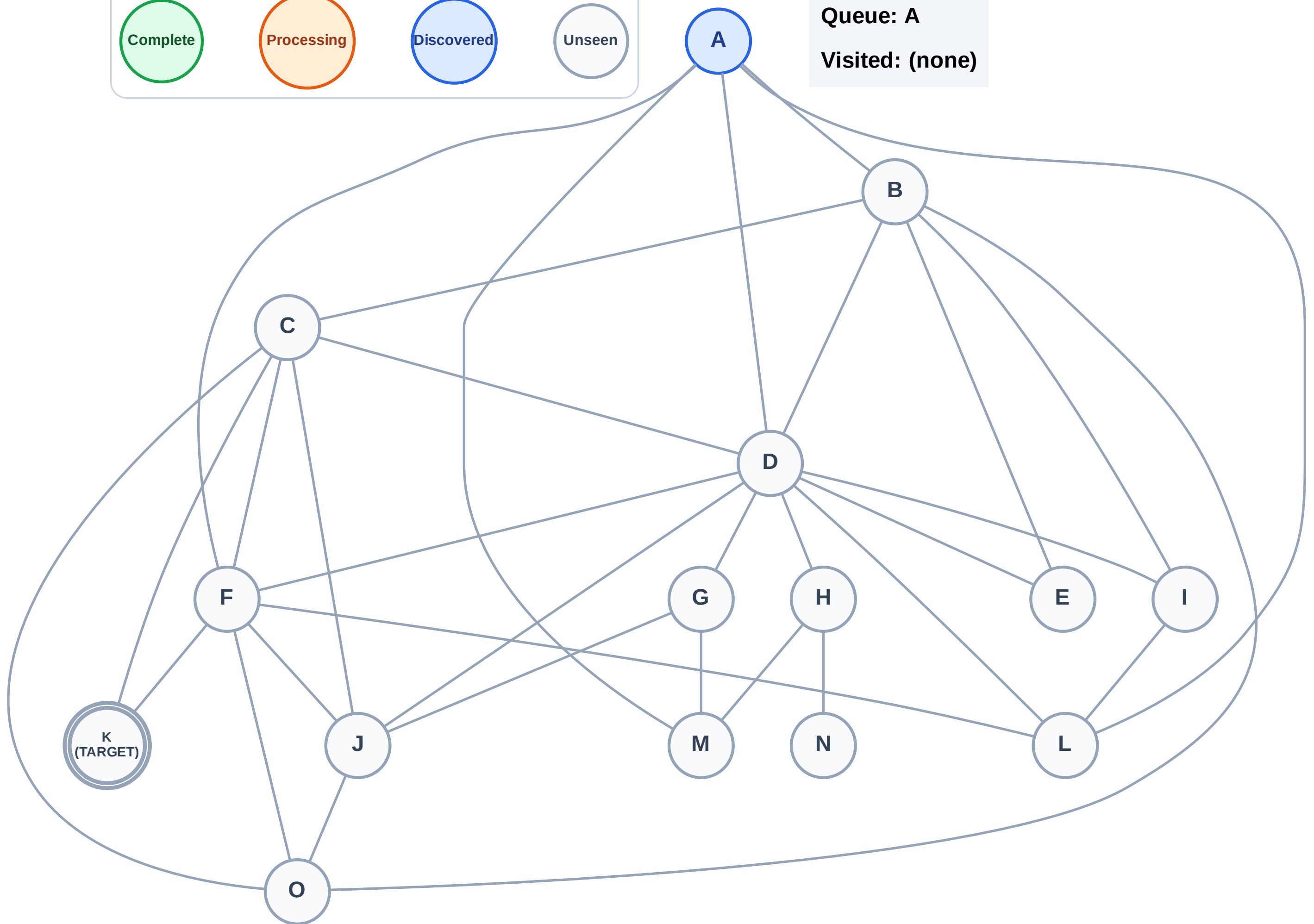
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Queue: A

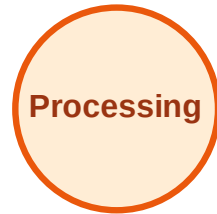
Visited: (none)



BFS Traversal

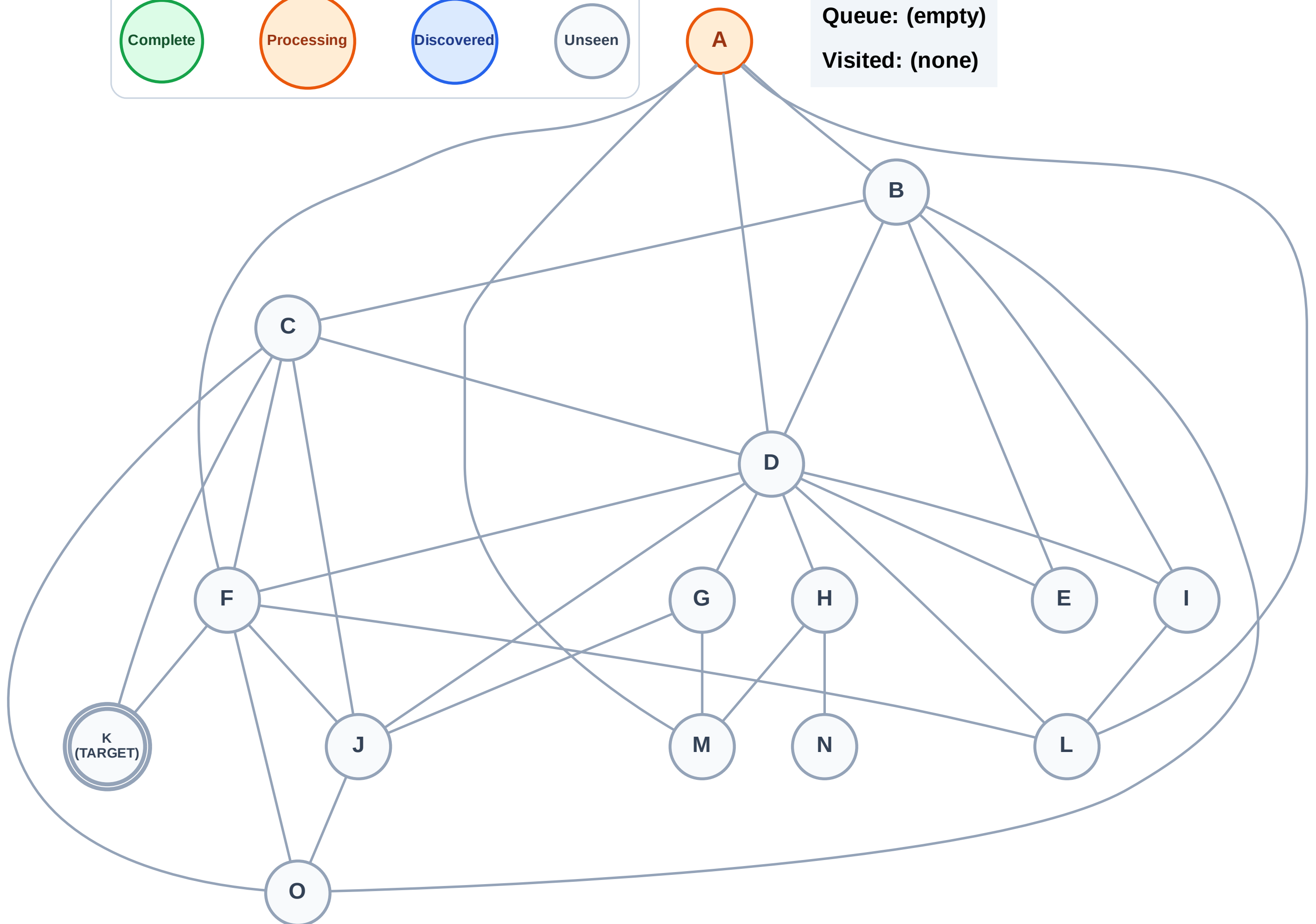
Step 2: Process node A

Legend



Queue: (empty)

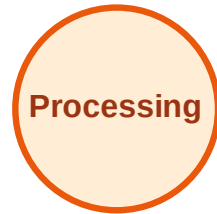
Visited: (none)



BFS Traversal

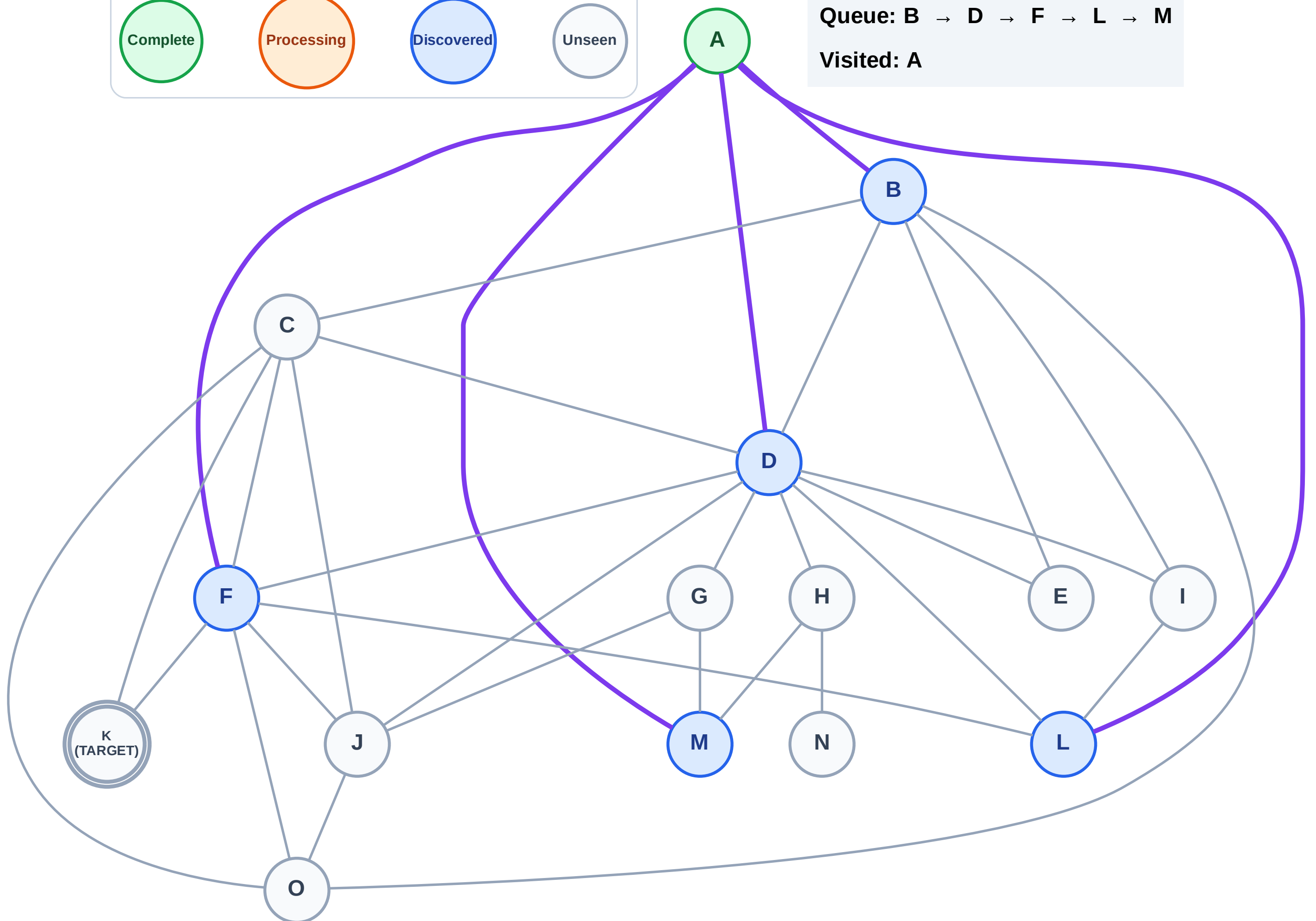
Step 3: Discover B, D, F, L, M from A

Legend



Queue: B → D → F → L → M

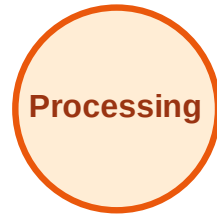
Visited: A



BFS Traversal

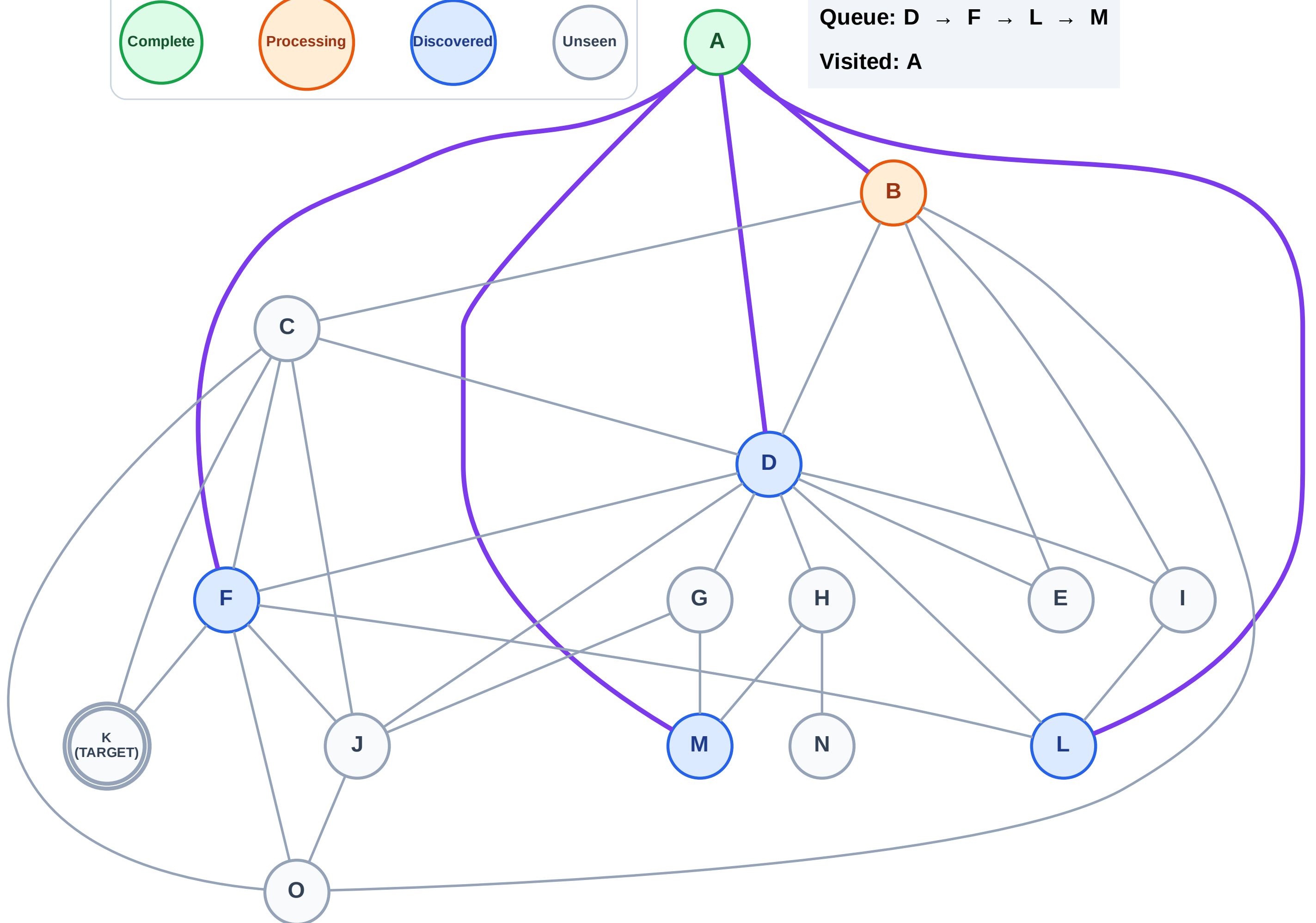
Step 4: Process node B

Legend



Queue: D → F → L → M

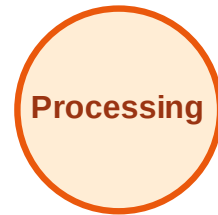
Visited: A



BFS Traversal

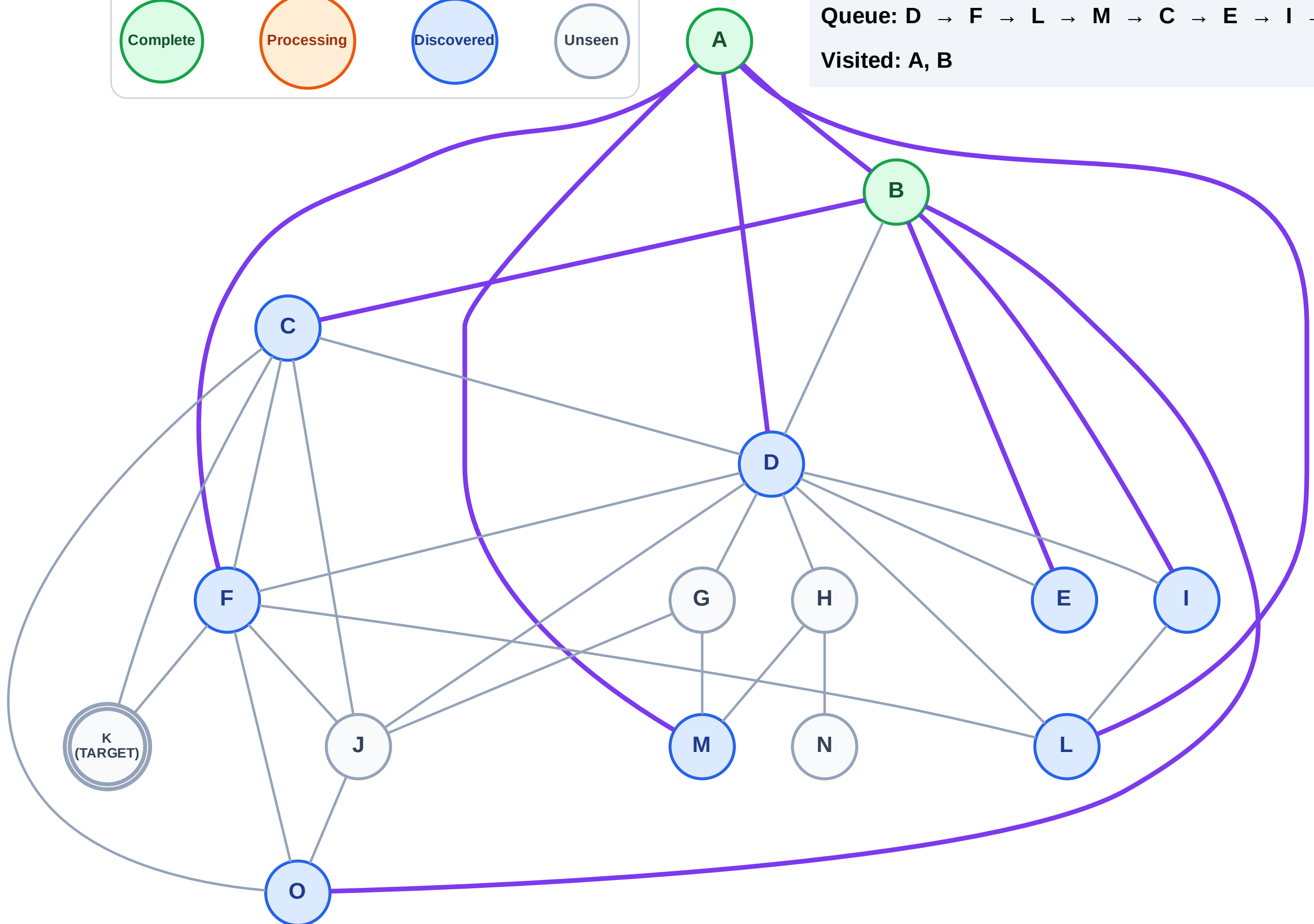
Step 5: Discover C, E, I, O from B

Legend



Queue: D → F → L → M → C → E → I → O

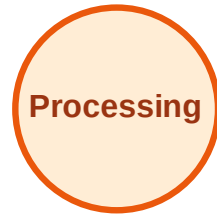
Visited: A, B



BFS Traversal

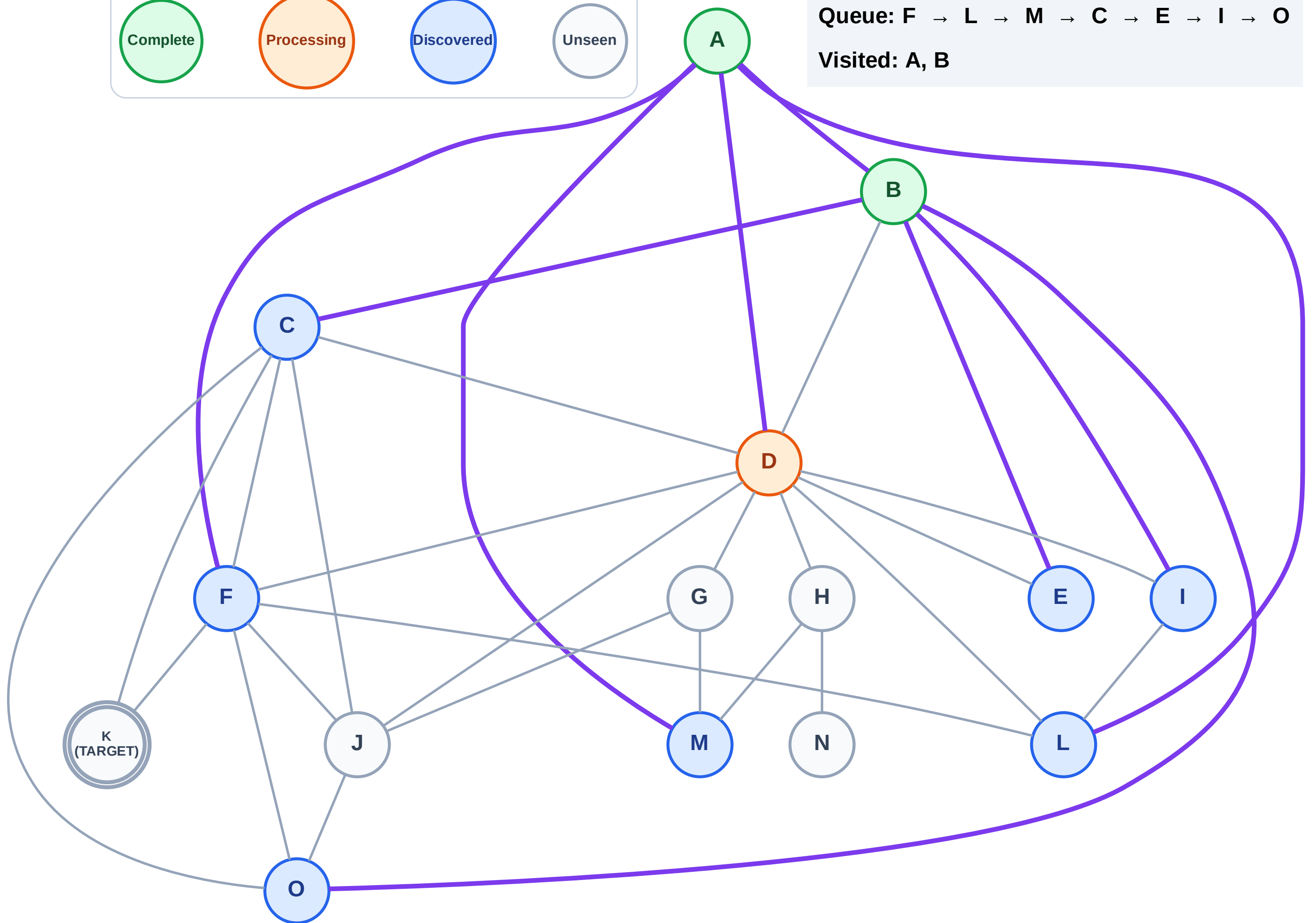
Step 6: Process node D

Legend



Queue: F → L → M → C → E → I → O

Visited: A, B



BFS Traversal

Step 7: Discover G, H, J from D

Legend

Complete

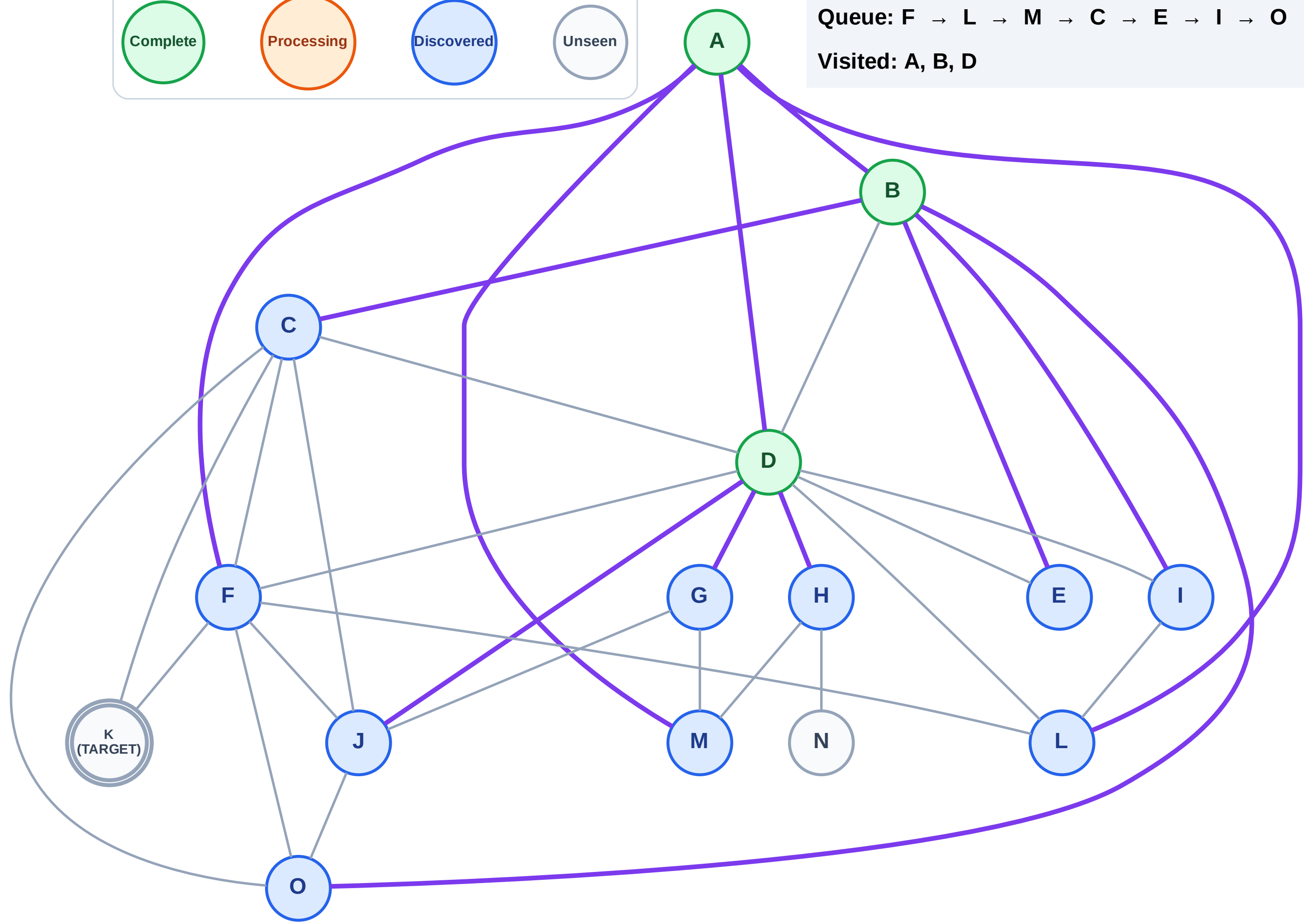
Processing

Discovered

Unseen

Queue: F → L → M → C → E → I → O → G → H → J

Visited: A, B, D



BFS Traversal

Step 8: Process node F

Legend

Complete

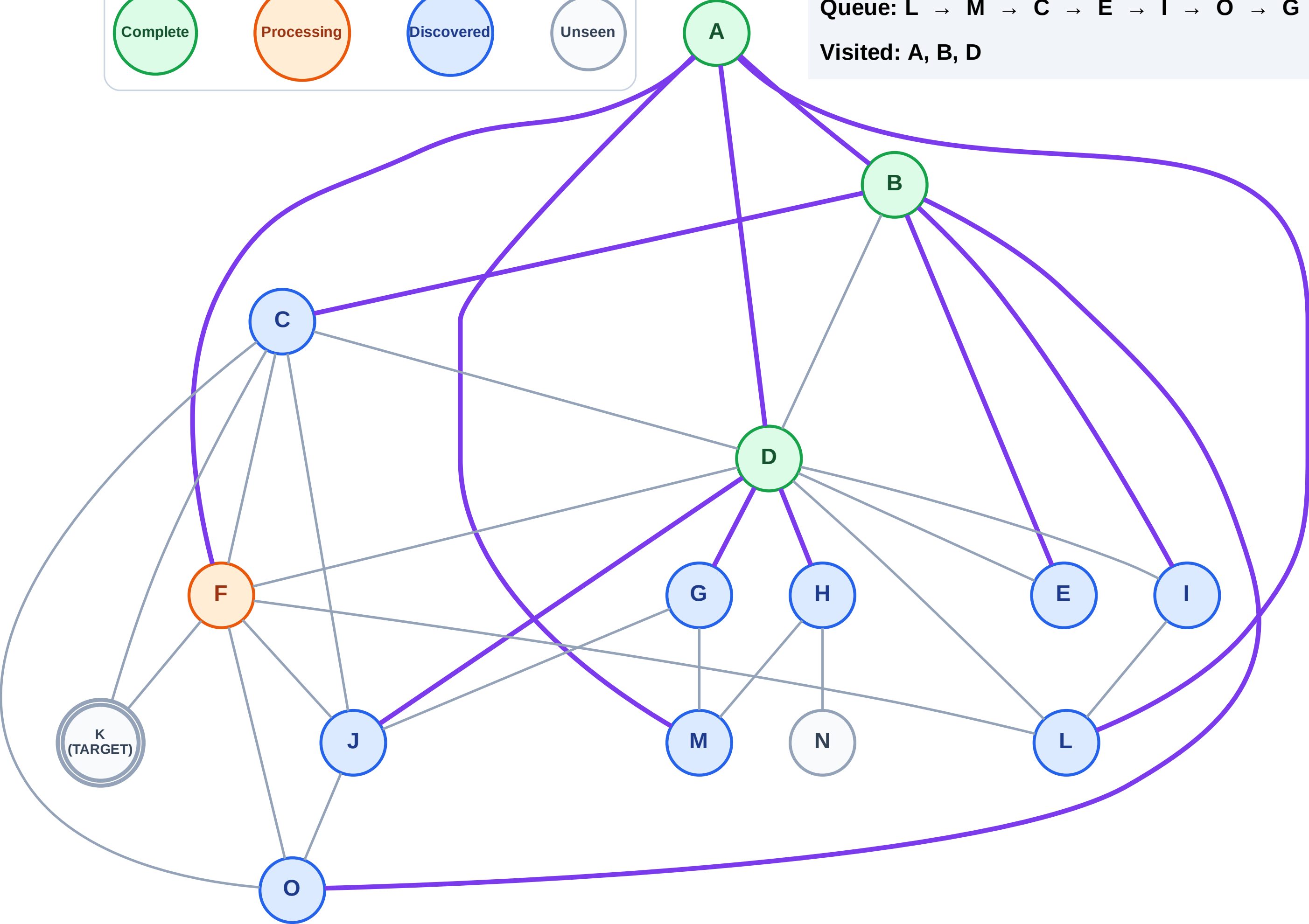
Processing

Discovered

Unseen

Queue: L → M → C → E → I → O → G → H → J

Visited: A, B, D



BFS Traversal

Step 9: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → F → K

Legend

Complete

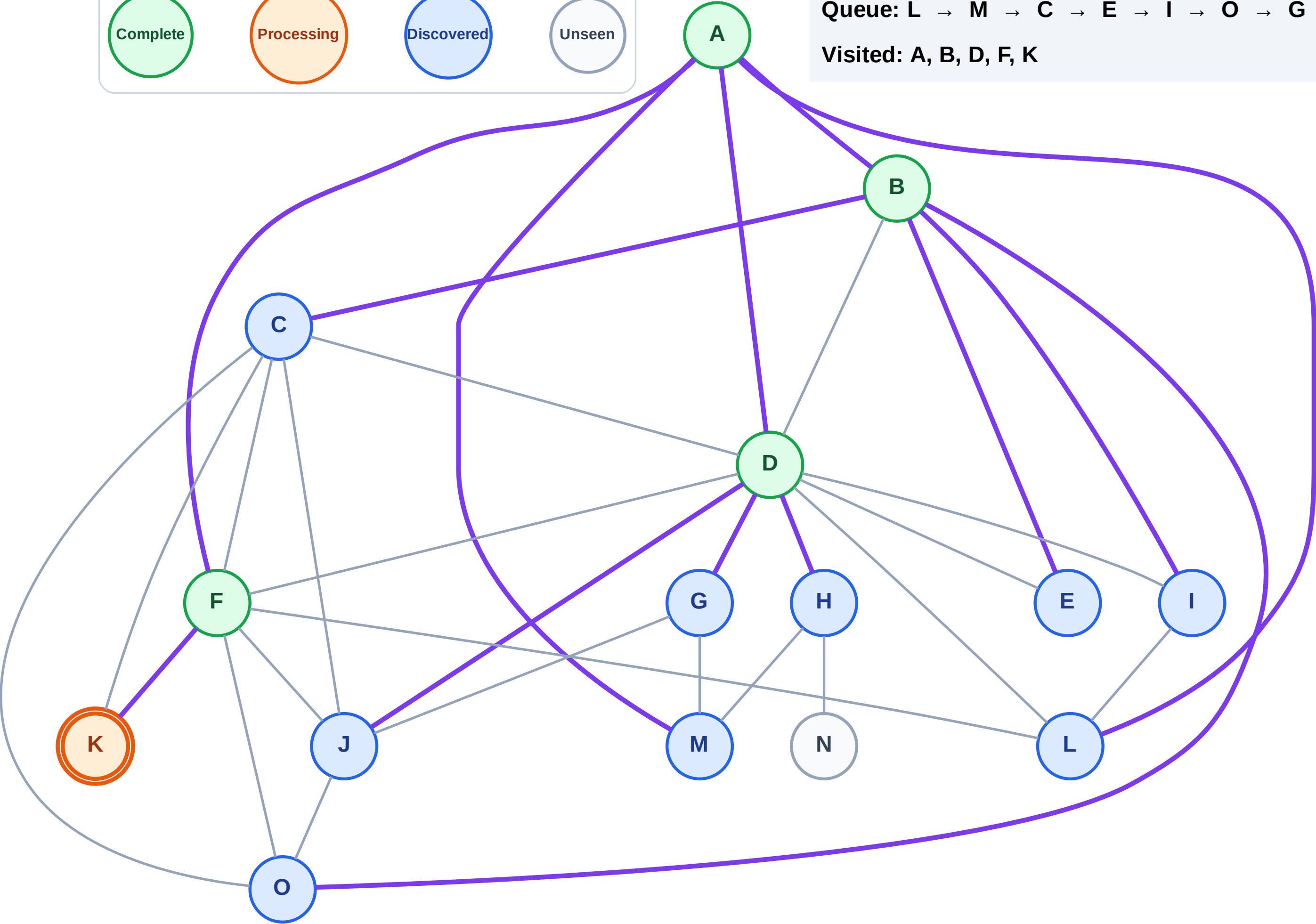
Processing

Discovered

Unseen

Queue: L → M → C → E → I → O → G → H → J

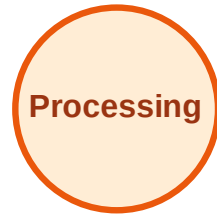
Visited: A, B, D, F, K



DFS Traversal

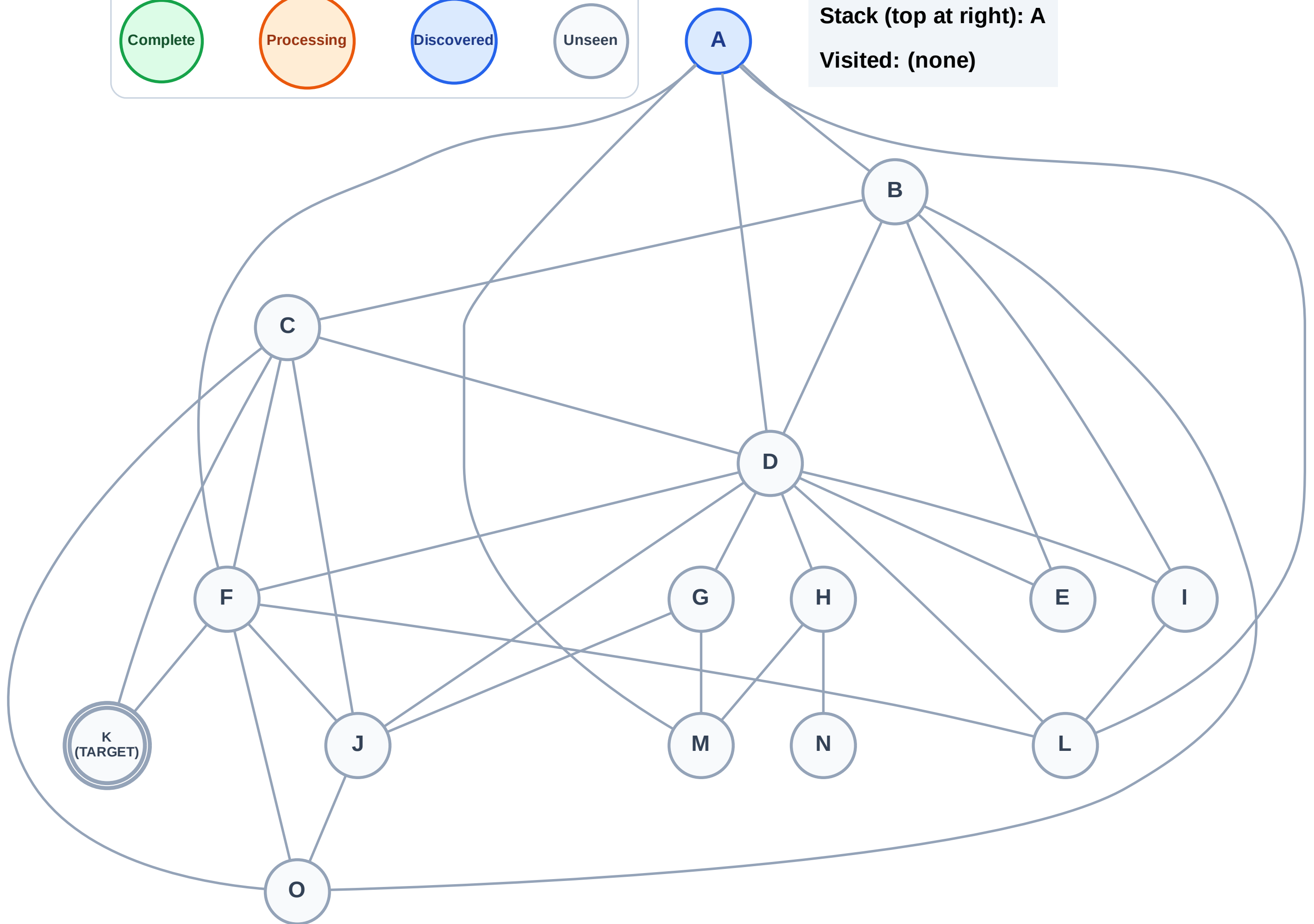
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Stack (top at right): A

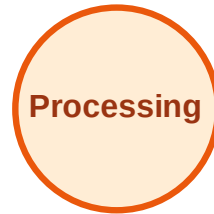
Visited: (none)



DFS Traversal

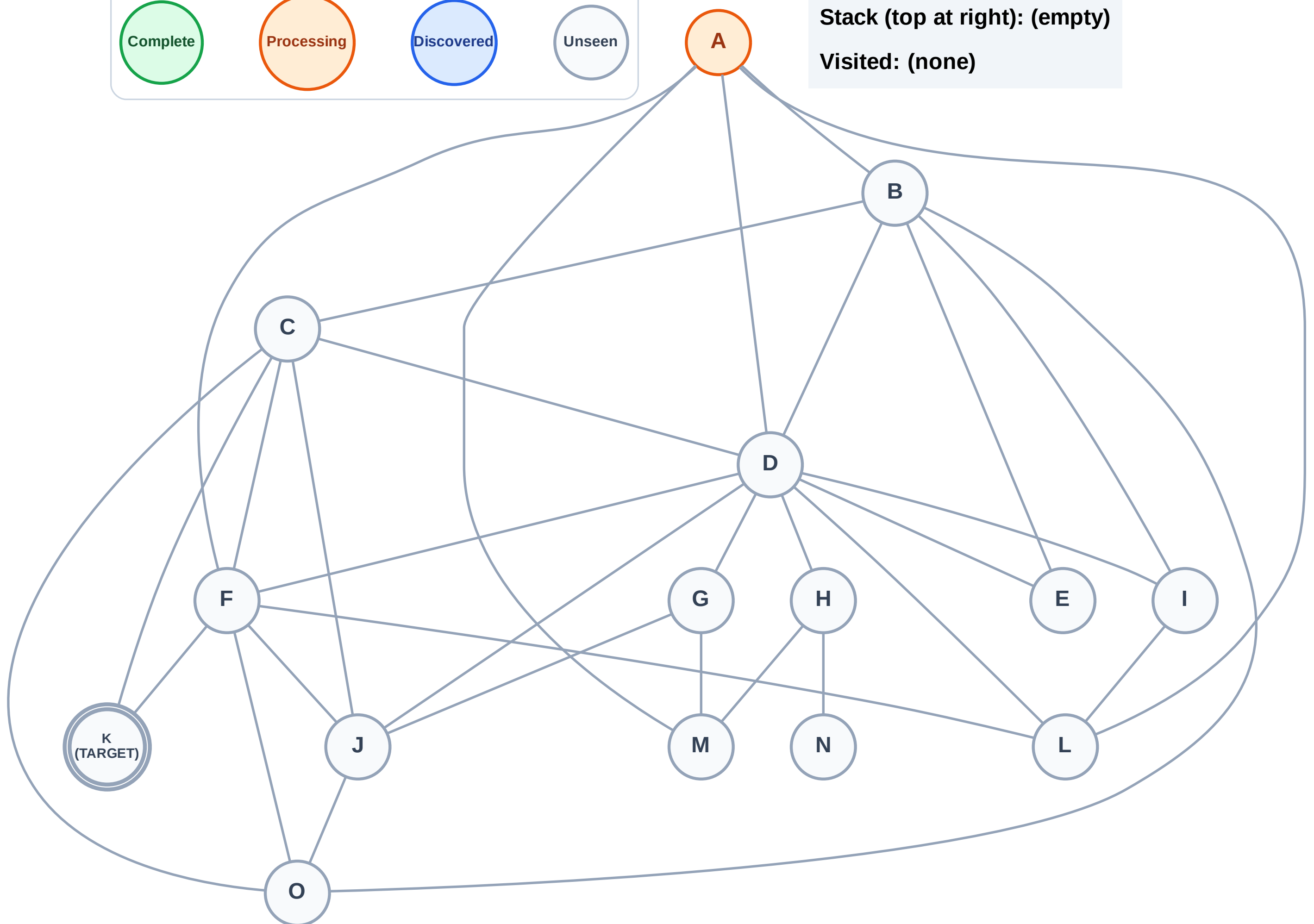
Step 2: Process node A

Legend



Stack (top at right): (empty)

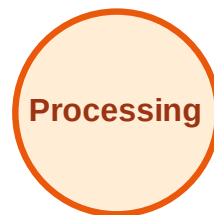
Visited: (none)



DFS Traversal

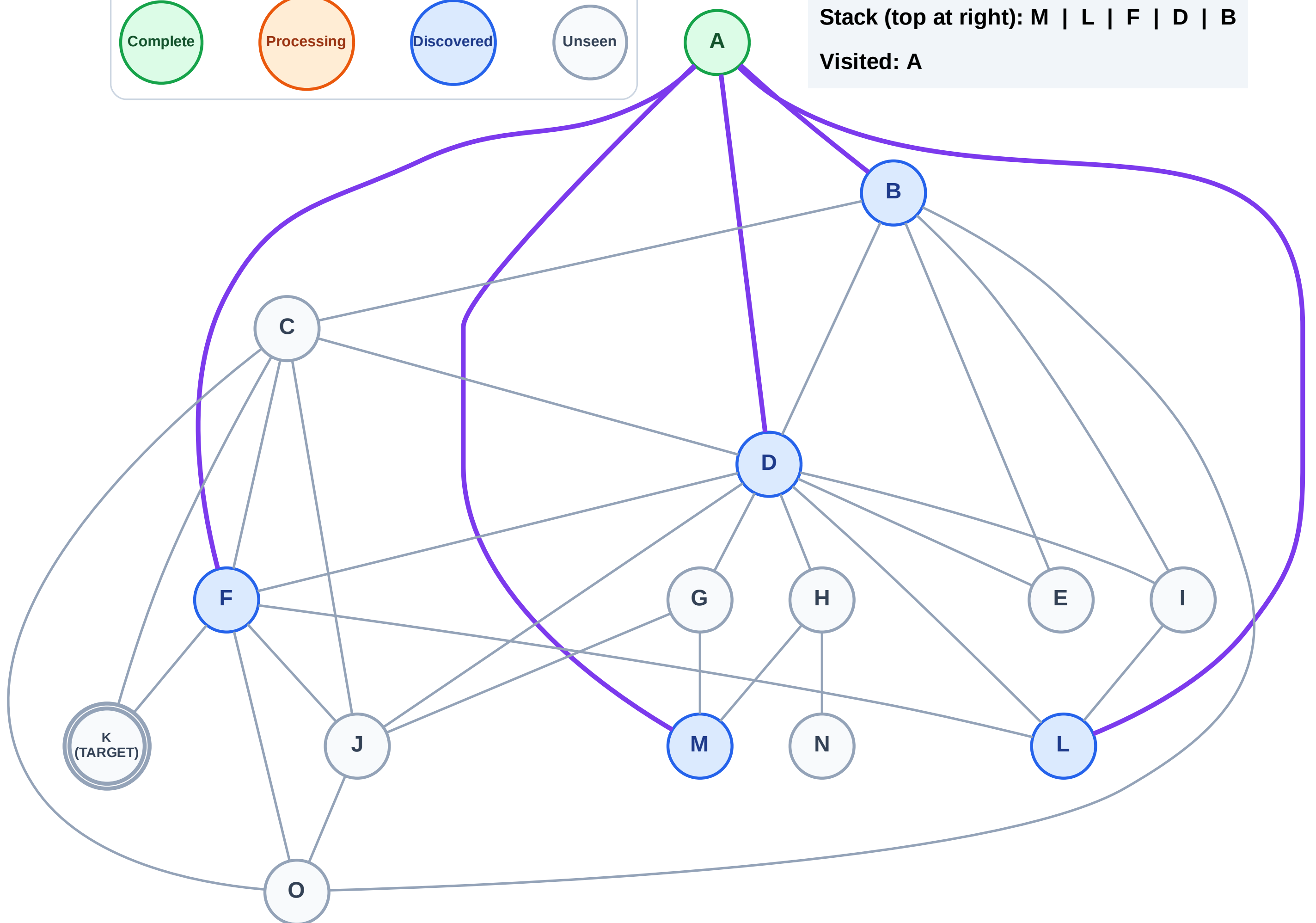
Step 3: Discover M, L, F, D, B from A

Legend



Stack (top at right): M | L | F | D | B

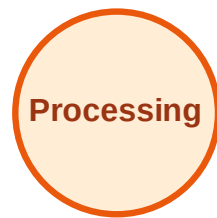
Visited: A



DFS Traversal

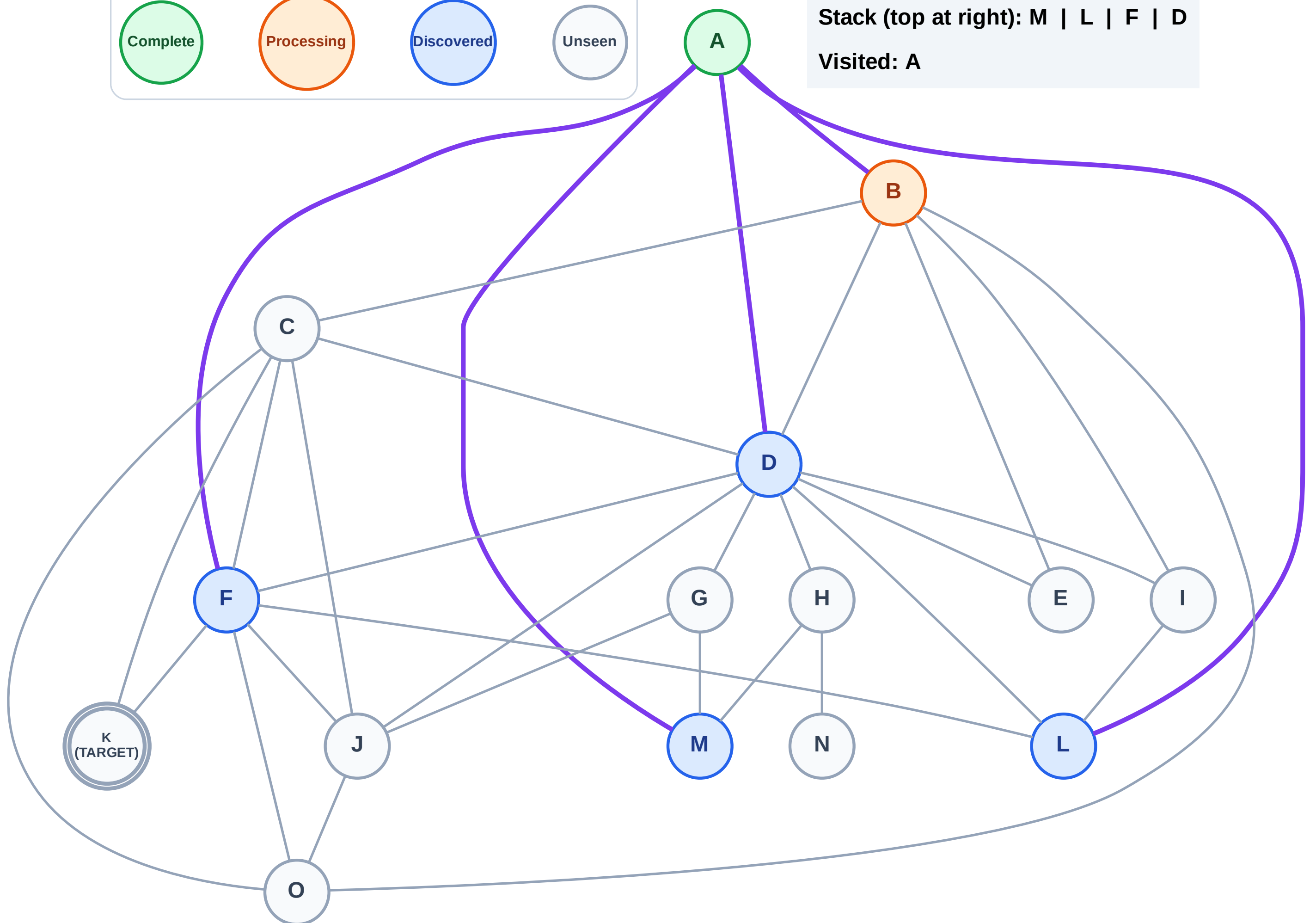
Step 4: Process node B

Legend



Stack (top at right): M | L | F | D

Visited: A



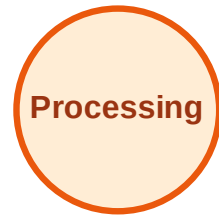
DFS Traversal

Step 5: Discover O, I, E, C from B

Legend



Complete



Processing



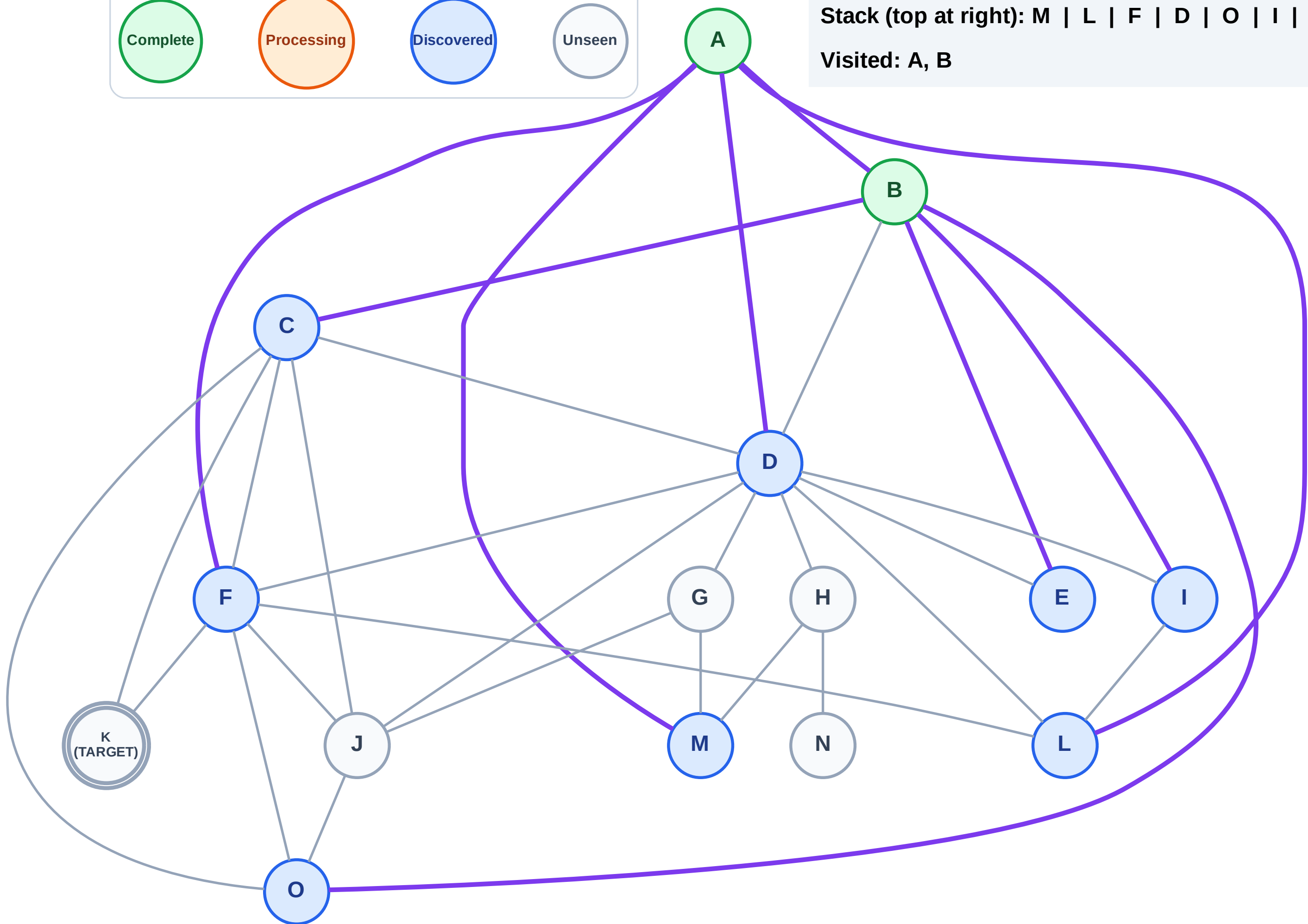
Discovered



Unseen

Stack (top at right): M | L | F | D | O | I | E | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

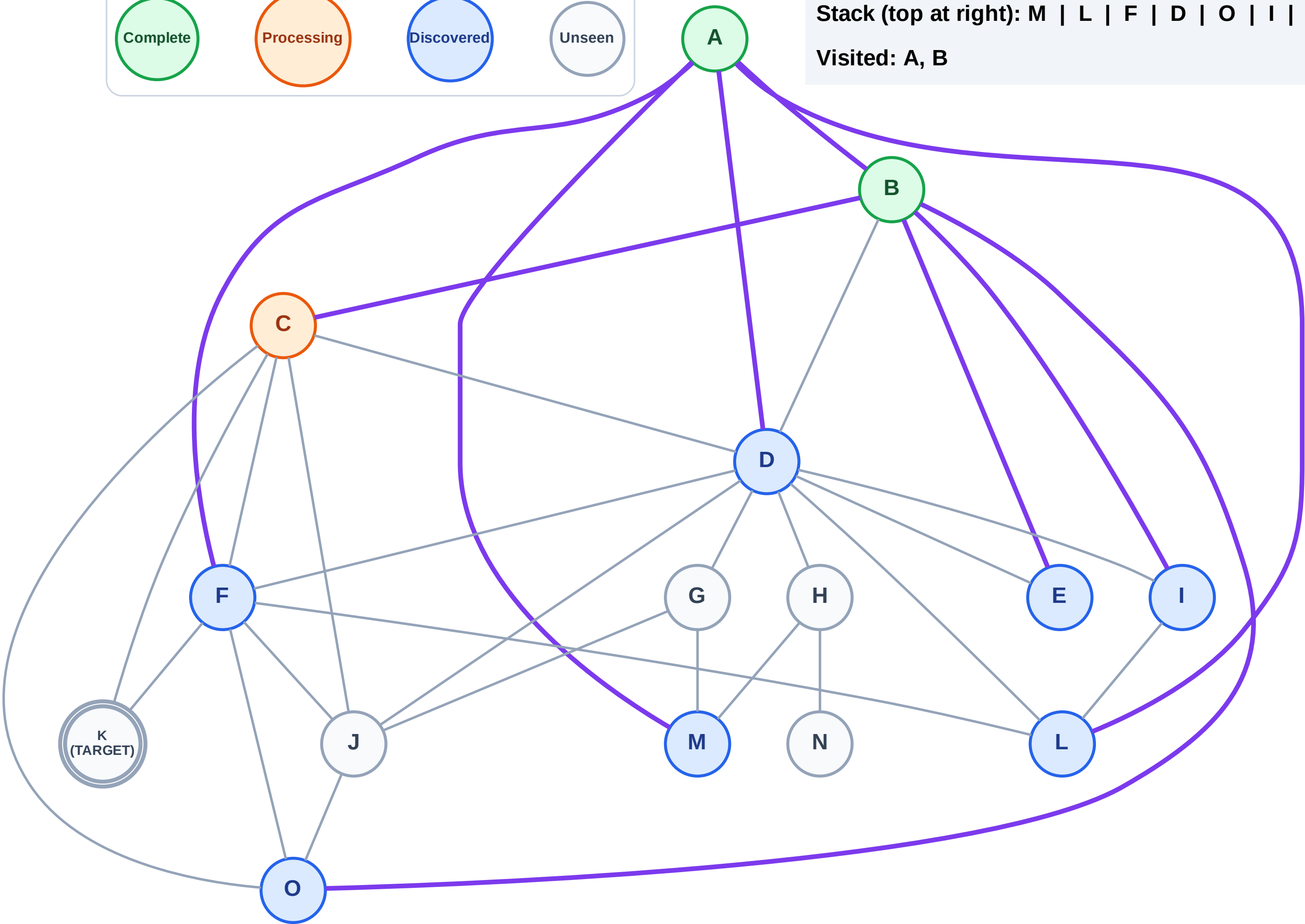
Processing

Discovered

Unseen

Stack (top at right): M | L | F | D | O | I | E

Visited: A, B



DFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → C → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): M | L | F | D | O | I | E
Visited: A, B, C, K

